

Assignment # 3

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Sec: D

Subject: Differential Equation

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Question # 1:

$$f(x) = -1 \quad (-1 < x < 0), \quad f(x) = 1 \quad (0 < x < 1)$$

$$P = 2L = 2$$

$$\text{Let } f(x) = \frac{1}{2} a_0 + \sum_{n=1}^{\infty} \left\{ a_n \cos\left(\frac{n\pi x}{L}\right) + b_n \sin\left(\frac{n\pi x}{L}\right) \right\}, \dots$$

$$a_0 = \frac{1}{L} \int_{-L}^L f(x) dx$$

$$a_0 = \frac{1}{1} \int_{-1}^0 (-1) dx + \int_0^1 (1) dx$$

$$a_0 = [-x]_{-1}^0 + [x]_0^1$$

$$a_0 = \{ (0) - (-1) \} + \{ (1) - (0) \}$$

$$a_0 = -1 + 1$$

$$\boxed{a_0 = 0}$$

$$a_n = \frac{1}{L} \int_{-L}^L f(x) \cos\left(\frac{n\pi x}{L}\right) dx$$

$$a_n = \frac{1}{1} \int_{-1}^0 (-1) \cos(n\pi x) dx + \int_0^1 (1) \cos(n\pi x) dx$$

$$a_n = -\left[\frac{\sin n\pi x}{n\pi}\right]_{-1}^0 + \left[\frac{\sin n\pi x}{n\pi}\right]_0^1$$

$$a_n = \frac{1}{n\pi} (0) + \frac{1}{n\pi} (\sin n\pi - 0)$$

$$\boxed{a_n = 0}$$

$$b_n = \frac{1}{L} \int_{-L}^L f(x) \sin\left(\frac{n\pi x}{L}\right) dx$$

$$b_n = \frac{1}{1} \int_{-1}^0 (-1) \sin(n\pi x) dx + \int_0^1 (1) \sin(n\pi x) dx$$

$$b_n = -\left[-\frac{\cos n\pi x}{n\pi}\right]_{-1}^0 + \left[-\frac{\cos n\pi x}{n\pi}\right]_0^1$$

$$b_n = \frac{1}{n\pi} \{\cos 0 - \cos n\pi\} - \frac{1}{n\pi} \{\cos n\pi - \cos 0\}$$

$$b_n = \frac{1}{n\pi} \{2 \cos 0 - 2 \cos n\pi\}$$

$$b_n = \frac{1}{n\pi} \{2 - 2(-1)^n\}$$

$$b_n = \frac{2}{n\pi} \{1 + (-1)^{n+1}\}$$

$$b_1 = \frac{2}{1 \cdot \pi} (2) = \frac{4}{\pi}, \quad b_2 = \frac{2}{2 \cdot \pi} (0) = 0, \quad b_3 = \frac{2}{3 \cdot \pi} (2) = \frac{4}{3\pi}$$

$$b_4 = 0, \quad b_5 = \frac{4}{5\pi}, \dots$$

So

$$f(x) = \frac{1}{2}(0) + \frac{4}{\pi} \sin \pi x + \frac{4}{3\pi} \sin 3\pi x + \frac{4}{5\pi} \sin 5\pi x + \dots$$

or

$$f(x) = \frac{4}{\pi} \left(\sin \pi x + \frac{1}{3} \sin 3\pi x + \frac{1}{5} \sin 5\pi x + \dots \right)$$

Question # 2:

$$f(x) = x \quad (0 < x < 2\pi)$$

$$\text{Let } x = \frac{1}{2} a_0 + a_1 \cos x + a_2 \cos 2x + \dots + b_1 \sin x + b_2 \sin 2x + \dots$$

By Euler's formulas:

$$a_0 = \frac{1}{\pi} \int_0^{2\pi} x \, dx$$

$$a_0 = \frac{1}{\pi} \left[\frac{x^2}{2} \right]_0^{2\pi}$$

$$a_0 = \frac{1}{2\pi} (4\pi^2 - 0)$$

$$\boxed{a_0 = 2\pi}$$

$$a_n = \frac{1}{\pi} \int_0^{2\pi} x \cos nx \, dx$$

$$a_n = \frac{1}{\pi} \left[x \left(\frac{\sin nx}{n} \right) \right]_0^{2\pi} - \frac{1}{\pi} \int_0^{2\pi} (1) \frac{\sin nx}{n}$$

$$a_n = \frac{1}{\pi} (0 - 0) - \frac{1}{\pi} \left[-\frac{\cos nx}{n^2} \right]_0^{2\pi}$$

$$a_n = \frac{1}{n^2\pi} (\cos 2nx - \cos 0)$$

$$\boxed{a_n = 0}$$

$$\therefore n = 1, 2, 3, \dots$$

$$a_1 = a_2 = a_3 = \dots = 0$$

Similarly,

$$b_n = \frac{1}{\pi} \int_0^{2\pi} x \sin nx \, dx$$

$$b_n = \frac{1}{\pi} \left[x \cdot \left(-\frac{\cos nx}{n} \right) \right]_0^{2\pi} - \frac{1}{\pi} \int_0^{2\pi} (1) \left(-\frac{\cos nx}{n} \right) dx$$

$$b_n = \frac{1}{\pi} \left(-\frac{2\pi}{n} \right) + \frac{1}{\pi} \left[\frac{\sin nx}{n^2} \right]_0^{2\pi}$$

$$\boxed{b_n = -\frac{2}{n}}$$

$$\therefore n = 1, 2, 3, \dots$$

$$b_1 = -2, \quad b_2 = -1, \quad b_3 = -\frac{2}{3}, \dots$$

Now Substituting the values of a 's and b 's

$$x = \frac{1}{2} (2\pi) - 2 \sin x - \sin 2x - \frac{2}{3} \sin 3x - \dots$$

Q8

$$x \div \pi - 2 \left(\sin x + \frac{1}{2} \sin 2x + \frac{1}{3} \sin 3x + \dots \right)$$

Ans